

Lecture Assignment 1: Formulating A data science task Credit Card Fraud Detection

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1 Introduction

For this lecture assignment, a dataset is required to be chosen to analyze data for the purposes of studying Data Science. The chosen subject to study for the assignment is Credit Card Fraud. We set out to analyze what in a credit card fraud dataset indicates fraud for the purposes of developing tools to better detect and prevent fraud in online transaction systems.

2 Motivation and Practical Relevance

Fraud can be a significant problem for companies and customers. If an information leak occurs or someone obtains information about another person's credit card, they can make purchases without the owner's permission.

An automated fraud detection system can identify suspicious transactions quickly, reducing the time and cost associated with manual review. Its primary users would include banks, credit card companies, and law-enforcement agencies. When the system flags a transaction, the financial institution can block it or contact the customer for confirmation.

The growing volume and speed of online payments make manual monitoring increasingly impractical. Automated systems can continuously apply mathematical algorithms to identify unusual patterns in customers' purchasing histories. Because digital payments have become central to commerce, particularly in Sweden, effective fraud detection is essential for maintaining the security of and public trust in the financial system.

3 Data Requirements and Collection Challenges

Fraud detection requires historical transaction data that describes when, where, and how each card was used. Relevant features may include the transaction time, location, amount, purchase type, and payment method. Information about a customer's typical spending behaviour is also valuable because unusual activity, such as an unexpected increase in transaction frequency, may indicate fraud.

The dataset must also label each transaction as legitimate or fraudulent. These labels allow a supervised machine-learning model to learn the differences between normal spending patterns and potentially fraudulent behaviour.

Collecting such data is challenging because financial transactions contain private and sensitive information. Banks and payment companies must comply with privacy and security regulations when storing, processing, or sharing customer data. Personal information may therefore need to be removed or anonymised before the data can be used for research.

Another challenge is class imbalance: fraudulent transactions are much less common than legitimate ones. As a result, obtaining enough representative examples of fraud can be difficult. Labels may also be delayed or inaccurate because suspicious transactions often require investigation or confirmation from customers before they can be classified as fraudulent.

4 Data Selection and Relevance

We choose [1] as our dataset. The dataset contains 284,807 credit card transactions, of which 492 are fraudulent. The Dataset contains information on time, amount, and 28 anonymised variables called V1 to V28. These variables were transformed using PCA, so that their original meaning is hidden. The set also has a variable called Class, which indicates whether the transaction is fraudulent. This dataset is very useful because it contains information on both fraud and normal transactions and can be used to train a machine learning model to detect possible fraud.

References

- [1] kaggle.com. *Credit Card Fraud Detection*. Last accessed 11 september 2026. 2021.